SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of ArtificialIntelligence and Machine Learning

ASSESSMENT1 – Analytical Problem solving (5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO2	**Assessment:**	Assessment 1: Analytical Problem solving (5 Questions × 10 Marks)
Weightage:	35% of CIA	**Total Marks:**	
CO2 Statement:	***Apply image enhancement, filtering, a techniques to analyze images, detect ed boundaries, and solve real-world Comp problems.***		
Student Name:	**S.Venkateswarlu**	**Reg. No.:**	**192425340**
Section / Batch:	**2024**	**Date:**	**21/10/2026**

Code of Conduct

I **S.Venkateswarlu /192425340** *(NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: **S.Venkateswarlu**

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Analytical Problems

Q1. A grayscale image contains the following pixel intensity values: 10, 10, 20, 20, 20, 30, 30, and 40. Construct the histogram by calculating the frequency of each intensity value. Calculate the cumulative distribution function (CDF) for the given image. Apply histogram equalization using the formula s = round((L − 1) × CDF), where L = 256. Determine the new intensity values and compare the contrast distribution before and after equalization.

Q2. A 3 × 3 image has the intensity values [[10, 20, 30], [20, 100, 20], [10, 20, 30]]. Apply a 3 × 3 averaging filter to calculate the new intensity value of the centre pixel. Replace the centre pixel with the calculated average while keeping the other pixels unchanged. Apply a median filter to the same neighbourhood and calculate the centre pixel value. Compare the results of the averaging and median filters and determine which filter better preserves image details in the presence of noise.

Q3. Consider a grayscale image with the intensity matrix [[50, 50, 50], [50, 100, 50], [50, 50, 50]]. Apply the Laplacian filter mask [[0, −1, 0], [−1, 4, −1], [0, −1, 0]] to calculate the response at the centre pixel. Use the calculated response to sharpen the centre pixel by adding it to the original intensity value. Determine the resulting centre pixel intensity after sharpening. Explain how the Laplacian filter enhances intensity transitions and fine details in the image.

Q4. A grayscale image contains the pixel intensity values [[40, 60, 120], [80, 150, 200], [30, 100, 180]]. Apply global thresholding with a threshold value of 100, assigning 1 to pixels greater than or equal to the threshold and 0 to the remaining pixels. Construct the resulting binary image. Calculate the number of foreground and background pixels after thresholding. Explain how the selected threshold separates objects from the background and identify one limitation of this method.

Q5. A grayscale image has the intensity matrix [[10, 10, 10], [10, 50, 90], [10, 50, 90]]. Apply the Sobel operator using the masks Gx = [[−1, 0, 1], [−2, 0, 2], [−1, 0, 1]] and Gy = [[−1, −2, −1], [0, 0, 0], [1, 2, 1]] to calculate the gradient components at the centre pixel. Calculate the gradient magnitude using $G = \sqrt{(Gx^2 + Gy^2)}$. Determine the dominant edge direction based on the calculated gradient components. Explain how the detected edge information can be used for boundary detection and image segmentation.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator Program Director
Signature:	Signature: Signature:
Date:	Date: Date:

Q1. Histogram Equalization

Given pixel values: 10, 10, 20, 20, 20, 30, 30, 40

Step 1: Calculate the histogram

Intensity (\(r\))	Frequency	Probability
10	2	2/8 = 0.25
20	3	3/8 = 0.375
30	2	2/8 = 0.25
40	1	1/8 = 0.125
Total	8	1.0

Step 2: Calculate the CDF

CDF is the cumulative sum of probabilities.

Intensity	Probability	CDF
10	0.25	0.25
20	0.375	0.625
30	0.25	0.875
40	0.125	1.0

Step 3: Apply histogram equalization

Formula:

\[s=\operatorname{round}((L-1)\times CDF) \]

Given $L=256$, so $L-1=255$.

- For intensity 10: $s=\operatorname{round}(255\times0.25)=64$
- For intensity 20: $s=\operatorname{round}(255\times0.625)=159$
- For intensity 30: $s=\operatorname{round}(255\times0.875)=223$
- For intensity 40: $s=\operatorname{round}(255\times1)=255$

Step 4: New intensity values

Original values:

10, 10, 20, 20, 20, 30, 30, 40

Equalized values:

64, 64, 159, 159, 159, 223, 223, 255

Conclusion: Histogram equalization spreads the pixel intensities across a wider range, improving contrast and making image details easier to distinguish.

Q2. Averaging Filter and Median Filter

Given image:

$$\begin{bmatrix} 10&20&30\\ 20&100&20\\ 10&20&30 \end{bmatrix}$$

Step 1: Apply the averaging filter

Formula:

$$\text{Average}=\frac{\text{Sum of all 9 pixels}}{9}$$

$$=\frac{10+20+30+20+100+20+10+20+30}{9}$$

$$=\frac{260}{9}=28.89$$

Therefore, the new centre pixel is approximately 29.

Filtered image:

$$\begin{bmatrix} 10&20&30\\ 20&29&20\\ 10&20&30 \end{bmatrix}$$

Step 2: Apply the median filter

Arrange all pixels in ascending order:

10, 10, 20, 20, 20, 20, 30, 30, 100

The middle (5th) value is 20.

Filtered image:

$$\begin{bmatrix} 10&20&30\\ 20&20&20\\ 10&20&30 \end{bmatrix}$$

Step 3: Compare the filters

Averaging filter	**Median filter**
Centre value = 29	Centre value = 20
Replaces the pixel with the mean	Replaces the pixel with the middle value
Can blur image details	Better preserves edges
Sensitive to extreme noise	Effective against salt-and-pepper noise

Conclusion: The median filter better preserves image details in the presence of impulse noise because the extreme value 100 does not strongly affect the median.

Q3. Laplacian Filter for Image Sharpening

Given image:

$$\begin{bmatrix} 50&50&50\\ 50&100&50\\ 50&50&50 \end{bmatrix}$$

Laplacian mask:

$$\begin{bmatrix} 0&-1&0\\ -1&4&-1\\ 0&-1&0 \end{bmatrix}$$

Step 1: Calculate the Laplacian response

Multiply corresponding pixels and add:

$$R=(4\times100)-(50+50+50+50)$$

$$R=400-200=\boxed{200}$$

Step 2: Sharpen the centre pixel

Given formula:

$$\text{New intensity}=\text{Original intensity}+R$$

$$=100+200=\boxed{300}$$

Since an 8-bit grayscale image has a maximum intensity of 255, the stored pixel value is clipped to 255.

Step 3: Explanation

- The Laplacian filter detects rapid intensity changes.

- It highlights edges and fine image details.
- Adding the response sharpens bright transitions under the given mask and formula.
- The response can amplify noise, so sharpening must be applied carefully.

Conclusion: The calculated centre intensity is 300 before clipping and 255 after clipping to the valid 8-bit range.

Q4. Global Thresholding

Given image:

$$\begin{bmatrix} 40&60&120\\ 80&150&200\\ 30&100&180 \end{bmatrix}$$

Threshold value: $T=100$

Rule:

$$g(x,y)= \begin{cases} 1,& f(x,y)\geq100\\ 0,& f(x,y)<100 \end{cases}$$

Step 1: Apply the threshold

- $40 \rightarrow 0$
- $60 \rightarrow 0$
- $120 \rightarrow 1$
- $80 \rightarrow 0$
- $150 \rightarrow 1$
- $200 \rightarrow 1$
- $30 \rightarrow 0$
- $100 \rightarrow 1$
- $180 \rightarrow 1$

Step 2: Resulting binary image

$$\boxed{ \begin{bmatrix} 0&0&1\\ 0&1&1\\ 0&1&1 \end{bmatrix}}$$

Step 3: Count the pixels

Pixel type	Count
Foreground (1)	5
Background (0)	4

Total 9

Step 4: Explanation

Global thresholding separates pixels into two groups based on a single threshold value.

- Foreground: Intensities greater than or equal to 100.
- Background: Intensities below 100.

Limitation: It may fail when lighting is uneven or when foreground and background have similar intensity values.

Conclusion: The binary image contains 5 foreground pixels and 4 background pixels.

Q5. Sobel Operator for Edge Detection

Given image:

$$\begin{bmatrix} 10&10&10\\ 10&50&90\\ 10&50&90 \end{bmatrix}$$

Sobel masks:

$$G_x= \begin{bmatrix} -1&0&1\\ -2&0&2\\ -1&0&1 \end{bmatrix}$$

$$G_y= \begin{bmatrix} -1&-2&-1\\ 0&0&0\\ 1&2&1 \end{bmatrix}$$

Step 1: Calculate G_x

Multiply the image and the G_x mask element by element, then add:

$$G_x=(-10+10)+(-20+180)+(-10+90)$$

$$G_x=0+160+80=\boxed{240}$$

Step 2: Calculate G_y

$$G_y=(-10-20-10)+(10+100+90)$$

$$G_y=-40+200=\boxed{160}$$

Step 3: Calculate gradient magnitude

Formula:

$$G=\sqrt{G_x^2+G_y^2}$$

$$G=\sqrt{240^2+160^2}$$

$$G=\sqrt{57600+25600}$$

$$G=\sqrt{83200}\approx\boxed{288.44}$$

Step 4: Determine the dominant edge direction

The gradient direction is:

$$\theta=\tan^{-1}\left(\frac{G_y}{G_x}\right)$$

$$\theta=\tan^{-1}\left(\frac{160}{240}\right)$$

$$\theta\approx\boxed{33.69^\circ}$$

The gradient points approximately 33.69° from the positive horizontal axis. Since G_x is greater than G_y, the intensity change is stronger in the horizontal direction. The edge itself is approximately perpendicular to the gradient, at 123.69° (or equivalently -56.31°).

Step 5: Applications

- Sobel detects boundaries between regions with different intensities.
- Strong gradient magnitudes indicate likely edges.
- Edge information helps identify object boundaries.
- It supports image segmentation by separating objects from their surroundings.

Conclusion: $G_x=240$, $G_y=160$, gradient magnitude $G\approx288.44$, and gradient direction $\theta\approx33.69^\circ$. The Sobel operator is useful for detecting edges and segmenting images.